.NET Program Execution Process

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.NET Program Execution Process

In this article, I will discuss the **.NET Program Execution Process Flow** in Detail. Please read our previous article discussing the Common Language Runtime (CLR) Architecture in detail. As .NET Developers, we should know when we create an application, how the application is compiled and how the .NET Framework executes the application. But before understanding the **.NET Program Execution** process, let us first understand how non-dot net applications such as C, VB, and C++ programs are executed.

Non-DOT NET Program Execution Process:

We know that computers only understand machine-level code. The Machine-level code is also known as native code or binary code. So when we compile a C, VB6, or C++ program, the language compiler compiles the language source code. It generates the native machine code (also called binary code), which can be understood by the underlying operating system. The general flow of the process is shown in the below image.

The Native code or machine code that is generated by the respective language compiler is specific to the operating system on which it is generated. If we take the compiled native code and try to run it on another operating system, then it will fail. So the problem with this type of program execution is that it is not portable from one platform to another. That means it is platform-dependent.

.NET Program Execution Process:

Now let us understand the .NET Program Execution process in detail. Using .NET, we can create different types of applications such as Console, Windows, Web, and Mobile Applications, irrespective of the type of application, when we execute any .NET application, the following things happen in order.

The .NET Application Source Code gets compiled into Microsoft Intermediate Language (MSIL), which is also called Intermediate language (IL) or Common Intermediate language (CIL) code. Both .NET and Non-DOT.NET applications generate an assembly when we compile the application. Generally, the assemblies have an extension of .DLL or .EXE based on the type of application we compile. For example, if we compile a Windows or Console application in .NET, we get an assembly of type EXE, whereas when we compile a Web or Class Library Project in .NET, we get an assembly of type .DLL.

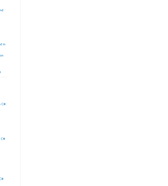
The difference between a .NET and NON-DOTNET assemblies is that .NET Assembly is an Intermediate Language format, whereas the NON-DOTNET assembly is a native format.

The NON .NET applications can run directly on top of the operating system as the NON-DOTNET assembly contains the native code, whereas .NET applications run on top of a virtual environment called **Common Language Runtime (CLR)**. It contains a component called **Just-In-Time Compiler (JIT)** which will convert the Intermediate language into native code, which can be understood by the underlying operating system.

.NET Program Execution Steps:

In .NET, the application execution consists of 2 steps. They are as follows.

In the step1 the respective language compiler compiles the Source Code into Intermediate Language (IL). In the 2nd step, the JIT compiler (CLR) will convert the Intermediate Language (IL) code into native code (Machine Code or Binary Code), which the underlying operating system can then execute. The above process is shown in the image below.



As the .NET assembly is in Intermediate Language (IL) format and not in native or machine code, the .NET assemblies are portable to any platform as long as the target platform has the **Common Language Runtime (CLR)**. The target platform's CLR converts the Intermediate Language code into native or machine code that the underlying operating system can understand.

Intermediate Language code is also called managed code. This is because CLR manages the code that runs inside it. For example, it is VB program, the developer is responsible for de-allocating the memory consumed by an object. If a developer forgets to de-allocate the memory, then it will result in memory leaks and will slow down the PC. But managed code does not need to worry about de-allocating the memory consumed by an object. Automatic memory management also known as garbage collection is provided by CLR. Apart from garbage collection, the CLR provides several other benefits, which we will discuss in a later session. Since CLR manages and executes the Intermediate Language (IL), it is also called the managed code.

.NET supports different programming languages like C#, VB, JIT and C++. C#, VB, and JIT can only generate managed code (IL), whereas C++ can generate both managed code (IL) and unmanaged code (Native code).

The native code is not stored permanently anywhere after we close the program the native code is thrown away. When we execute the program again, the native code is generated again.

The .NET program is similar to the Java program's execution. In Java, we have bytecodes and JVM (Java Virtual Machine) whereas in .NET we have Intermediate Language and CLR (Common Language Runtime).

In the next article, I will discuss the **Intermediate Language (IL) Code** in C# with examples. In this article, I try to explain the DOT.NET Program Execution process. I hope you enjoy the article.

Dot Net Tutorials
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Pranaya Rout has published more than 3000+ articles over the last 11 year career. Pranaya Rout has very good experience with Microsoft Technologies including C#, VB, ASP.NET MVC, ASP.NET Web Forms, .NET Core, SQL Server, .NET Core, LINQ, EF Core, WPF/SL, Oracle, ASP.NET Core, Cloud Computing, Microservices, Design Patterns and still learning new technologies.

Previous Lesson
Common Language Runtime in .NET Framework

Next Lesson
Intermediate Language (CIL)/(IL) & (L)ANG Code in C#

5 thoughts on "NET Program Execution Process"

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OCT 9, 2019 AT 12:40 AM
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